

Guidelines for writing the project report

Time Series

2024 – 2025

1. Introduction:

Contextualization and purpose of the work, brief description of the methodology used.

2. Data and exploratory analysis/data transformation

3. Model proposals (SARIMA type and/or ETS model and/or GARCH model):

Box & Jenkins methodology if SARIMA models; this includes model identification, parameter estimation, diagnostic evaluation (parameters correlation, parameters significance, residual analysis).

4. Future observations forecast:

Includes the analysis of its accuracy, confidence intervals (if possible), use of bootstrap methodology (if necessary). Accuracy of the forecasts should be compared.

5. Results, discussion/conclusions:

Discussion should include the model choice.

6. References

7. Appendix:

Auxiliary/not so important results are included in the Appendix.

NOTE: These are some guidelines - it is not an exhaustive list, but it contains what is most important. It can be organized in other way, if more suitable.

Do not forget: you should analyze all your data to model the particular model classes that are suitable to the data trajectory behavior, then you have to split the data into a training set (for model purposes) and a test set (for forecast purposes).

Deadline: Report and the correspondent script should be sent by the 16th of June 2025.

Presentation: On the 18th of June 2025 (Final Exam day).